

Assignment:- 5 unit:- 4

Name: Kaushal Kumar Raj

PRN: 240205241023

Branch: B.Tech CSE with AI & ML

Semester: IV

Subject: Deep Learning

① What is Convolutional Neural Network (CNN)?

→ A convolutional neural network (CNN) is a type of deep learning model specifically designed for processing image and spatial data.

It uses convolution layers, pooling layers, and fully connected layers to automatically learn patterns such as edges, textures and objects from images.

② Why are CNNs preferred over fully connected networks for image data?

→ CNNs are preferred because they:

- captures spatial relationship between pixels.
- use fewer parameters due to weight sharing.
- Automatically extract features from images
- Require less computation compared to fully connected networks.

③ What is a convolutional layer?

→ A convolution layer is a fundamental layer of CNN that applies filters (kernels) to the input image to extract features.

Each filter slides across the image and performs element-wise multiplication to produce a feature map. It helps detect patterns like: edges, textures, shapes.

④ What is the purpose of pooling layers in CNNs?

→ Pooling layers are used to:

- Reduce the size of feature maps
- Decrease computation cost
- Prevent overfitting
- Make features invariant to small shifts in the image

Common Pooling methods include: max pooling, Average Pooling

⑤ What is a feature map?

→ A feature map is the output produced when a convolution filter is applied to an image.

It highlights specific patterns detected by that filter.

ex:

A filter designed to detect edges will generate an edge feature map.

⑥ what are the common applications of CNNs?

→ CNNs are widely used in:

- Image classification
- Object detection
- Facial recognition
- Medical image analysis
- Autonomous vehicles
- Video surveillance

⑦ what is LeNet and where was it used?

→ LeNet is one of the earliest convolutional neural networks developed by Yann LeCun in 1998. It was used for handwritten digit recognition, particularly for reading ZIP codes on postal envelopes.

⑧ what is AlexNet and why is it significant in deep learning history?

→ AlexNet is a deep CNN developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton. It became famous after winning the ImageNet Large Scale Visual Recognition Challenge in 2012.

Its success marked the breakthrough of deep learning in computer vision.

⑨ what is meant by very deep networks in CNNs?

→ very deep networks refer to CNN architectures with many layers (sometimes hundreds).

These networks learn complex hierarchical features from data and improve performance for large-scale image recognition tasks.

ex: architectures include VGGNet and ResNet.

⑩ what is a Residual Network (ResNet)?

→ A Residual Network (ResNet) is a deep CNN architecture that introduces skip connections (residual connections).

These connections allow the networks to bypass certain layers and help solve the vanishing gradient problem, making it possible to train very deep neural networks.

⑪ what is a Recurrent Neural Network (RNN)?

→ A Recurrent Network (RNN) is a neural network designed to process sequential data such as text, speech, or time-series data.

It maintains a hidden state (memory) that stores information from previous inputs.

⑫ why are RNNs suitable for sequential data?

→ RNNs are suitable because they:

- Remember previous info
- Capture temporal dependencies
- Process variable-length sequences

This makes them useful for language processing and time series prediction

⑬ what is Back propagation Through Time (BPTT)?

→ Back propagation Through Time (BPTT) is the training algorithm used for RNNs.

In this method, the RNN is unrolled across time steps, and errors are propagated backward through each time step to update the weights.

⑭ what are vanishing and exploding gradient problems?

→ During training of deep networks,

Vanishing Gradient

- Gradients become extremely small
- Learning slows or stops

Exploding Gradient

- Gradients become extremely large
- causes unstable learning.

⑮ what is the purpose of gated mechanisms in RNNs?

→ Gated mechanisms control the flow of info inside RNNs.

They help:

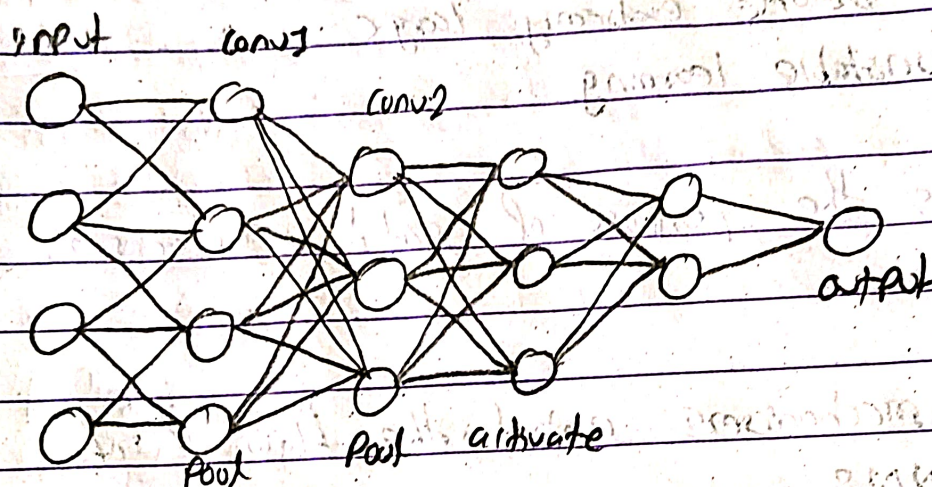
- Preserve long-term dependencies
- Avoid vanishing gradients
- select which information to keep or forget

ex: includes LSTM and GRU networks.

Long questions

① Explain the architecture of a CNN with convolution, Pooling and fully connected layers.

→ A CNN is a deep learning architecture designed for processing image data:



1) **input layer**: Receives the image data such as RGB image (height $\times$ width $\times$ channels)
ex: $224 \times 224 \times 3$

2) **convolution layer**: applies filters to input image
like: Detect edges, Detect textures, Detect shapes. Each feature produces a feature map.

3) **activation layer**
commonly uses ReLU (Rectified Linear unit).
use: introduces non-linearity
• Allows learning complex patterns.

4) Pooling layer

Pooling reduces the spatial size of feature maps.

Types:

- Max Pooling
- Average Pooling

Benefits:

- Reduces computation
- controls overfitting

5) Fully connected layer

→ The extracted features are flattened and connected to dense layers.

Purpose: perform classification.

CNN workflow

- i) Image input
- ii) Convolution layer
- iii) Activation (ReLU)
- iv) Pooling
- v) Fully connected layer
- vi) Output classification.

② Describe the working of convolution and pooling layers with suitable examples.

→ Convolution involves sliding a kernel (filter) across an image.

ex:

input matrix:

1 2 3

4 5 6

7 8 9

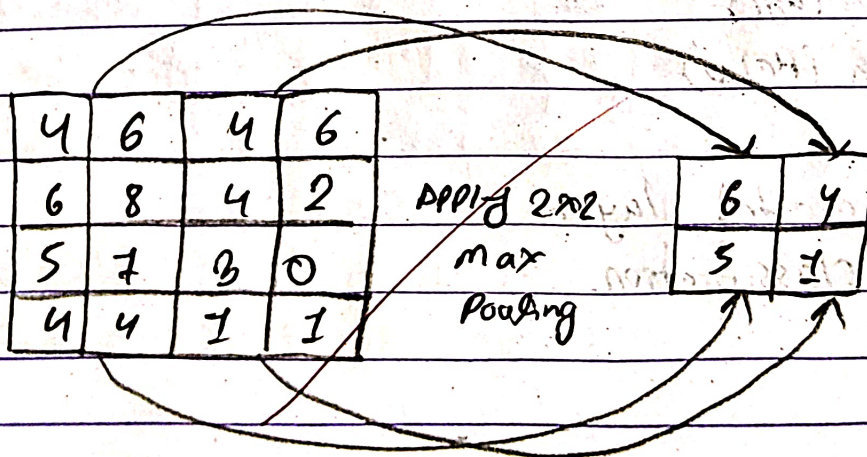
Filter:

1 0

0 1

The filter multiplies overlapping values and sums them to produce a value in feature map.

Pooling reduces feature map dimensions.



Pooling keeps important features while reducing computation.

③ Explain various application of CNNs in real-world problems.

→ CNNs are widely used in many fields

1) Image classification

- Classifies images into categorical such as animals, vehicles or objects

2) object detection

- Detects and locates objects in images.
ex: self-driving cars detecting pedestrians.

3) Face Recognition

- used in security systems and smartphones

4) medical Image Analysis

- CNNs detect disease from medical scans such as:
 - X-ray
 - MRI
 - CT Scans

5) Autonomous vehicles

- Helps identify roads, traffic signs, and obstacles.

④ Discuss the Let Net architecture and its importance.

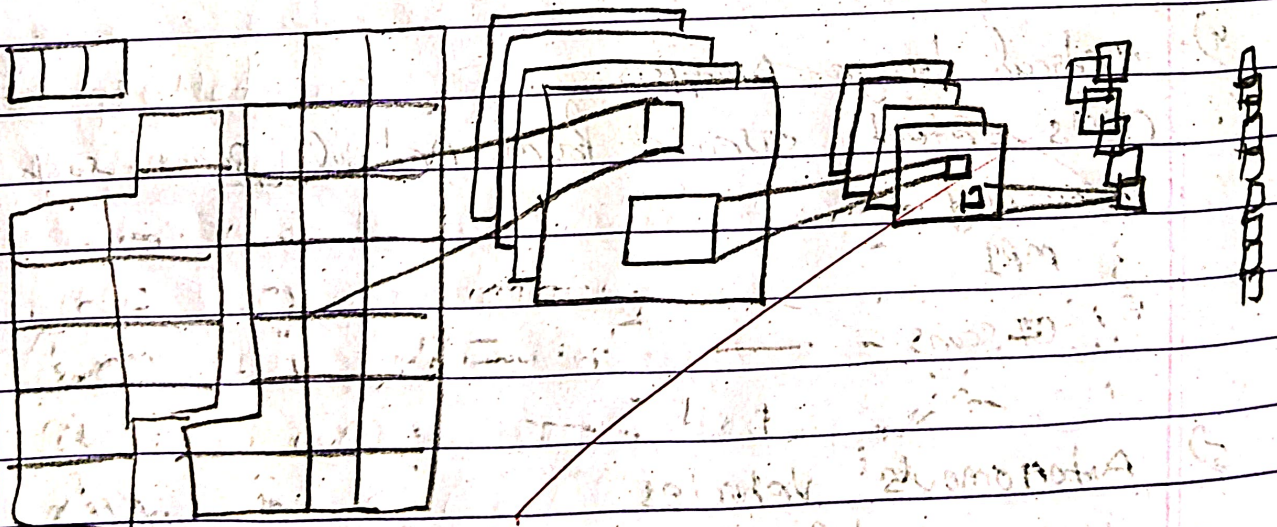
→ Let Net is one of the earliest CNN models developed by Yann LeCun.

Architecture.

- 1) Input layer
- 2) Convolution layer
- 3) Pooling layer
- 4) Convolution layer
- 5) Pooling layer
- 6) Fully connected layers
- 7) output layer

Importance

- First successful CNN architecture
- used for handwritten digit recognition
- Demonstrated automatic feature extraction



⑤ Explain the AlexNet architecture and highlight its contributions to deep learning.

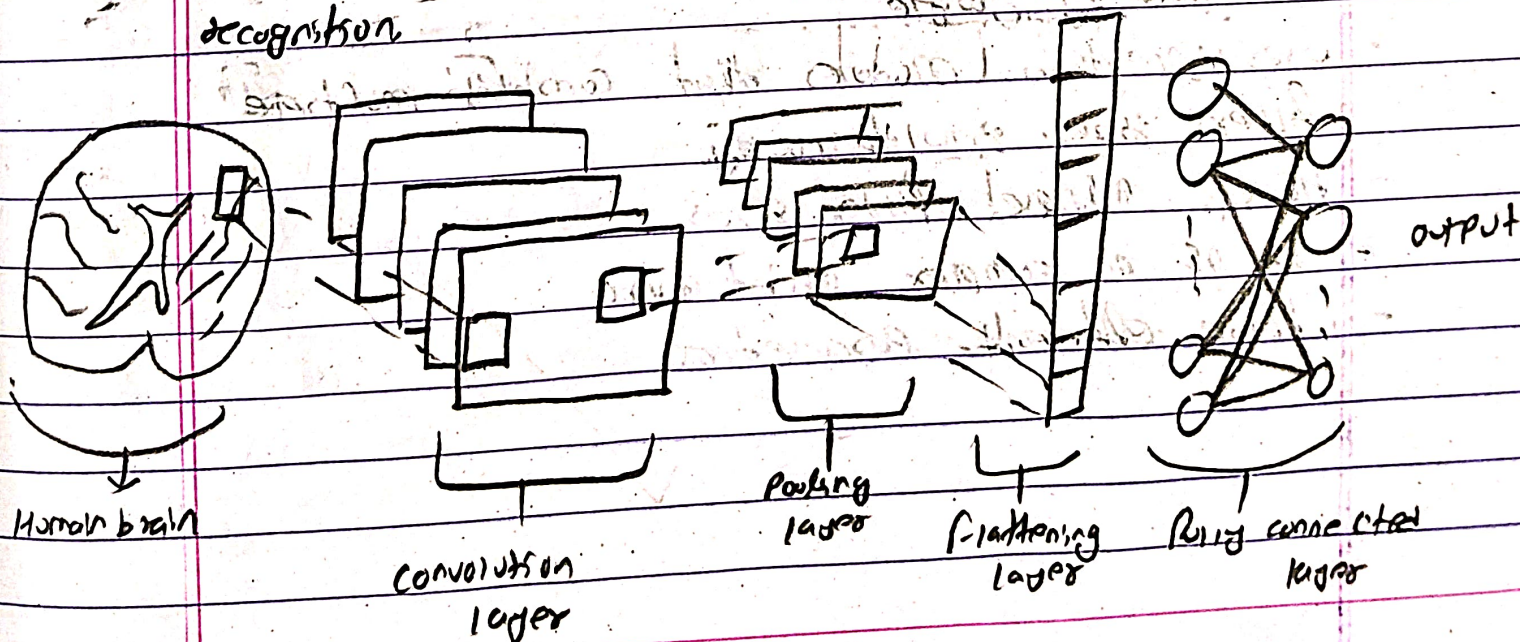
→ AlexNet was developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton.

Architecture:-

- 5 convolution layers
- 3 fully connected layers
- ReLU function
- Dropout regularization
- max pooling

Contributions:-

- Introduced ReLU activation
- used GPU training
- Applied dropout to reduce overfitting
- Achieved breakthrough accuracy in image recognition.



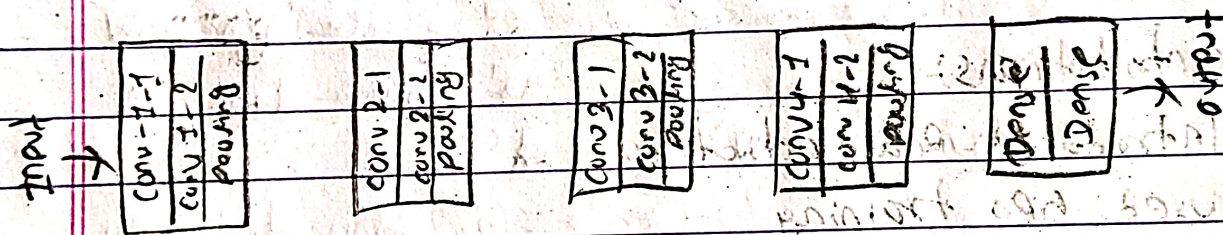
⑥ Explain VGGNet and GoogleNet architectures, comparing design philosophies.

→ VGGNet

- Developed by visual geometry group

Characteristics:-

- uses 3×3 convolution filters
- very deep networks (16 or 19 layers)
- simple architecture
- Very large parameters.



GoogleNet

Developed by Google

- uses inception modules that combine multiple filter sizes simultaneously.
- Have multiple filter sizes
- It is a complex architecture
- Have efficient parameters.

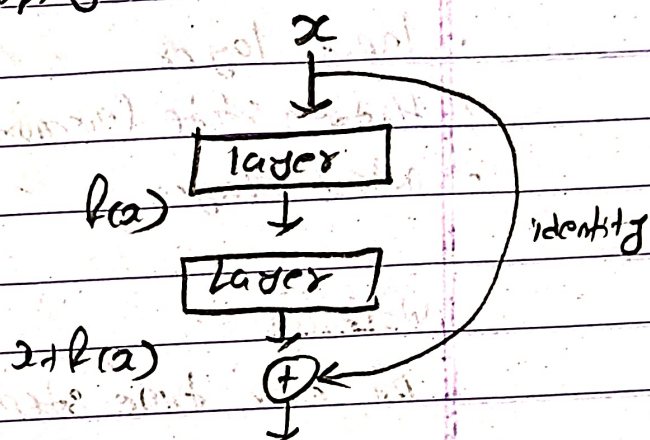
⑦ Explain the ResNet architecture and how residual connections solve training difficulties.

→ ResNet was introduced by Kaiming He.
Residual Learning

- Instead of learning direct mapping $H(x)$, ResNet learns residual mapping:

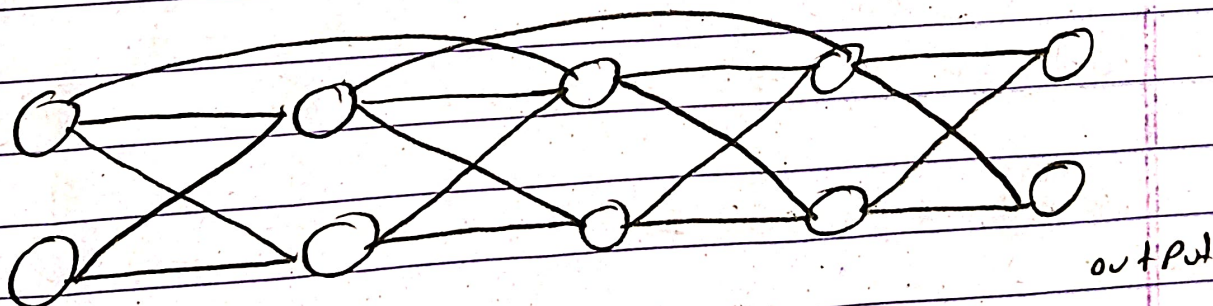
$$f(x) = H(x) - x$$

output: $f(x) + x$



Advantages:

- Solves vanishing gradient problem
- Enables very deep networks
- Improves training efficiency.



input

output

Fig: Deep Residual network.

⑧ Explain the architecture and working of RNNs.

→ A Recurrent Neural network (RNN) processes sequential data.

Components:

- Input layer
- Hidden state (memory)
- output layer

Working

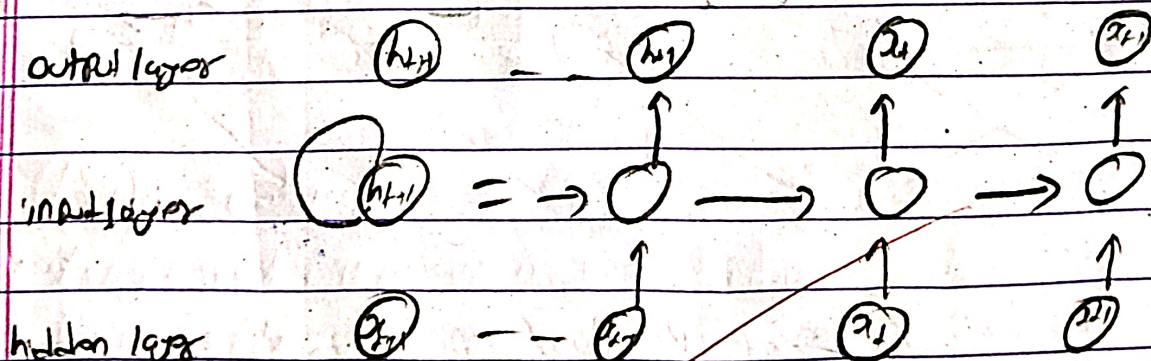
At each time step:

$$h_t = f(Wx_t + Uh_{t-1})$$

where,

h_t = hidden state

x_t = input



Q) What is Back propagation Through time (BPTT)? Explain its working in detail.

→ Back Propagation Through Time (BPTT) is the training algorithm used for RNNs.

Steps:-

- i) RNNs is unrolled over time steps
- ii) Forward pass computes outputs
- iii) Loss is calculated
- iv) Gradient propagate backward through time
- v) weights are updated.

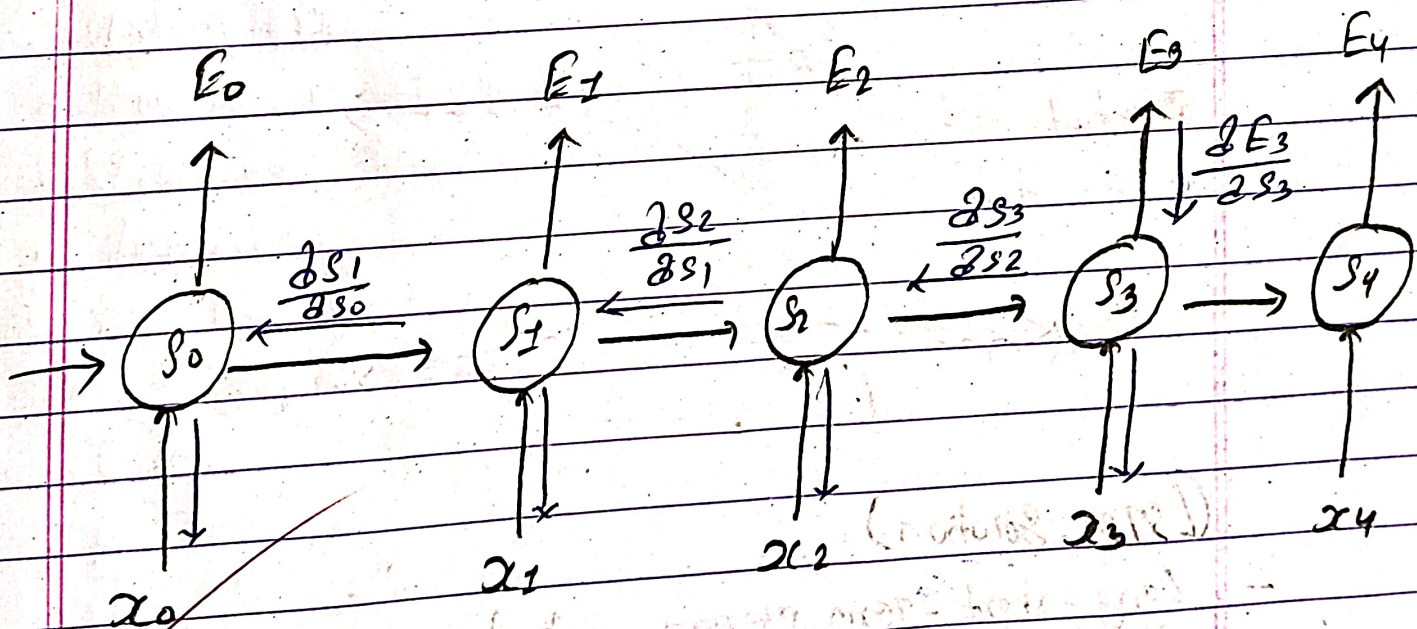


Fig: Backpropagation Through Time.

⑩ Explain the problems of vanishing and exploding gradients and how LSTM / GRU overcome them.

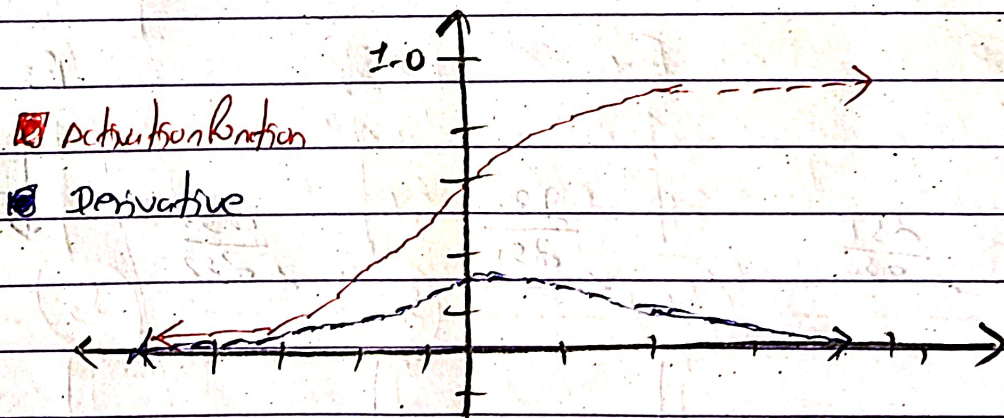
→ Vanishing gradient.

- Gradient becomes extremely small during backpropagation
- Result: Network fails to learn long-term dependencies.

Exploding gradients

→ Gradients becomes extremely large

Result: ~~that~~ Training becomes unstable.



(LSTM solution)

- Long-short-term memory introduces gates:
- forget gate
- input gate
- output gate

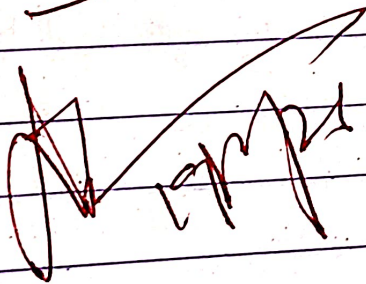
These gates regulate information flow.

GRU Solution

- Gated Recurrent unit (GRU) simplifies LSTM with:
 - update gate
 - reset gate

Advantages:

- Faster training
- Fewer parameters.

A large, stylized handwritten mark in red ink, possibly a signature or a large checkmark, located below the advantages list.